

Adam Najjar

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Summary of Qualifications

- Third-year Aerospace Engineering student at TMU with CAD, design, build and test experience from five engineering projects.
- Analytical problem-solving in CATIA V5, MATLAB, Simulink and OpenRocket, applied to a gearbox, a 6-DOF model and a rocket.
- Teamwork and communication skills from two 4-person design teams, including delivering the rocket team's presentation.
- Organization and attention to detail from standardizing 30,000+ product records across sales, technical and web teams.
- Leadership and mentoring skills from supervising 9 swim instructors and founding a debate program for 30 students.

Technical Skills

Design & CAD: CATIA V5 | SolidWorks (self-taught) | OpenRocket | GD&T | Mechanical design | Reverse engineering

Analysis & Software: MATLAB | Simulink | FlightGear | Swift | C | SQL | Excel | CATSY | Kinematic analysis | Test data analysis

Project & Build: Technical reports | Gantt charts | Cost estimates | Risk analysis | Decision matrices | 3D printing

Education

Bachelor of Engineering, Aerospace Engineering

Sep 2024 – Exp. Apr 2028

Toronto Metropolitan University (TMU), Toronto, ON

Relevant Courses: Engineering Design | Thermodynamics & Heat Transfer | Fluid Mechanics | Numerical Analysis

Academic & Personal Projects

Metropolitan Aerospace Rocket Society (MARS) | Toronto Metropolitan University, OpenRocket

Feb 2026 – May 2026

- Collaborated in a 4-person team to design, build and test a competition rocket, and delivered the team's design presentation.
- Simulated the design in OpenRocket, then analyzed onboard altitude, velocity and pressure telemetry against the prediction.
- Achieved a successful launch with full parachute deployment and recovery and no significant damage to the vehicle.

Walking Robot Mechanism Design | Course Project, MATLAB

Mar 2026 – Apr 2026

- Evaluated Hoekens, Klann and Jansen linkages in a 3-person team, co-writing MATLAB displacement and velocity analyses.
- Selected the Jansen linkage for its flat stance stroke, keeping 2 legs grounded; the robot walked the full 3 m course.

Flight Simulator Physics Engine | MATLAB, Simulink, FlightGear

Feb 2026 – Apr 2026

- Developed a 6-DOF aircraft dynamics model with aerodynamic, control-surface and actuator models for realistic flight.
- Linked the model to FlightGear and a joystick, enabling real-time visualization and pilot-in-the-loop testing of the model.

Robotic Gearbox Reverse Engineering | Course Project, CATIA V5

Jan 2026 – Feb 2026

- Reverse-engineered a competitor's robotic gearbox in a 4-person team, measuring and modelling all 11 parts with GD&T.
- Built team consensus on redesign trade-offs by scoring 14 concepts across 6 components in weighted decision matrices.
- Delivered an easier-to-assemble drop-in design with a 10% lighter spoked gear and one fewer part at the same gear ratio.

Airbus A380 Overhead Access System | Engineering Design Project, CATIA V5

Feb 2025 – Apr 2025

- Designed a retractable ceiling ladder in CATIA V5 to improve passenger access to overhead bins, sized for 150 kg loads.
- Evaluated gear and pulley drives in a 4-person team on safety, cost and manufacturability, selecting the pulley drive.
- Planned the project schedule and cost estimate, and documented design rationale and safety features in a technical report.

Work Experience

Marketing Associate | Techspan Industries — Oakville, ON

May 2023 – Present

- Organized 30,000+ product records into clear categories with descriptions, making products easier for customers to find.
- Applied attention to detail in Excel and CATSY to fix unclear names, missing info and wrong images, sharply cutting complaints.
- Communicate across sales, technical and web teams to agree on clear, consistent naming, contributing to higher site traffic.

Head Coach & Swim Instructor | Making Waves Swim School — Mississauga, ON

Jan 2023 – Jan 2025

- Demonstrated strong planning and organization leading daily operations and supervising 9 instructors teaching 27 swimmers.
- Built a standardized training program and mentored instructors with regular feedback, visibly improving their teaching.

Volunteer & Leadership

Outreach Lead | Yusr Funeral Financial Aid — Mississauga, ON

Oct 2024 – Present

- Built partnerships with 3 community organizations, helping grow the program past 100 participants and raise \$20,000+.

Volunteer | Islamic Relief Canada — Tuzla & Sarajevo, Bosnia

Jun 2024 – Jul 2024

- Ran daily activities for a group of 20 children, of 200 in the orphanage program, and raised \$9,000 in humanitarian aid.

Founder, Debate & Public Speaking Program | Seeds of Leadership — GTA, ON

Nov 2021 – Jun 2024

- Took the initiative to found a student debate program, running 30-student sessions and tournaments with guest speakers.